Viscous Drop Problem

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Problem Formulation

This problem is an extension of the work done by Gabbard et al. (2025), specifically the reduced-order kinematic match method.

Gabbard et al. (2025) considers the deformations of an axisymmetric droplet's rebound motion. To achieve this, they use a reduced order model, where the shape of the droplet is approximated by the sum of spherical harmonic modes. Each one of these modes have an amplitude, which determines the weight of the mode. The modes determining deformation amplitude are denoted Al (A of L). These amplitudes are found by solving a seconday differential equation in terms of Al.

The coefficients of these differential equations, are approximated in Gabbard et al. (2025). However, this has only been theoretically proven a good approximation for the regime Oh * Sqrt(L) < 0.03, where Oh is the Ohnesorge number and L is the mode number (Molacek & Bush (2012)).

This problem aims to extend the work of Gabbard et al. (2025) by solving for the true values of the coefficients, so that the resulting ODE yields more accurate values of Al, therefore improving the accuracy of the model for higher values of Oh, which represent more viscous cases.

Finding the true values of the coefficient requires us to solve a transcendental equation which has infinite roots, for the roots with smallest real-positive components. Additionally, this transcendental equation shifts only slightly for small differences in its parameters, allowing us to use a continuation method.

Non-dimensionalization and Rescaling

Gabbrd et al. (2025) Eq 3.4 shows the dimensional ODE of Al and Bl, with estimated coefficients. Ignoring the Bl term, this form should be analogous to Ragazzo (2019) Eq. 20, which shows the true coefficients. Aguero (2026) shows the full ODE with the Bl term incorporated as well.

$$\dot{\mathscr{U}}_l' = -\frac{\sigma l\,(l+2)\,(l-1)}{\rho R^3}\mathscr{A}_l' - 2(2l+1)(l-1)\frac{\mu}{\rho R^2}\mathscr{U}_l' - \frac{l}{\rho R}\mathscr{B}_l',$$

Gabbard (2025) Eq. 3.4

$$\ddot{A}_l = -\sigma_{al}\sigma_{bl}A_l - (\sigma_{al}+\sigma_{bl})\dot{A}_l$$

Ragazzo (2019) Eq. 20

$$\ddot{A}_l + (\tilde{\sigma}_a + \tilde{\sigma}_b)\dot{A}_l + \tilde{\sigma}_a\tilde{\sigma}_b A_l = -\frac{l}{\rho R}B_l$$

Aguero (2026)

$$\ddot{A}_l' + 2(2l+1)(l-1)\frac{\mu}{\rho R^2}\dot{A}_l' + l(l+2)(l-1)\frac{\sigma}{\rho R^3}A_l' + l\left(\frac{1}{\rho R}\right)B_l' = 0$$

$$\Big\downarrow \text{ is an approximation of}$$

$$\ddot{A}_l' + (\bar{\sigma}_a + \bar{\sigma}_b)\dot{A}_l' + (\bar{\sigma}_a\bar{\sigma}_b)A_l' + \frac{l}{\rho R}B_l' = 0$$

Gabbrd et al. (2025)'s problem formulation is Non-dimensionalized. In order to apply the current work onto their larger method, the equation from Aguero (2026) should be Non-dimensionalized as well, with the following scales from Gabbard et al. (2025).

Gabbard (2025) Characteristic Scales

Density i.t.o ρ

Length i.t.o R

time i.t.o $t_\sigma = \sqrt{\frac{\rho R^3}{\sigma}}$

Non-dimensionalization and Rescaling

Galbart et al, (2025) Eq 3.4 $\ddot{A}_l' + 2(2l+1)(l-1)\frac{\mu}{\rho R^2}\dot{A}_l' + l(l+2)(l-1)\frac{\sigma}{\rho R^3}A_l' + l\left(\frac{1}{\rho R}\right)B_l' = 0$

$\downarrow$ is an approximation of

Galbart et al, (2025) Eq 3.16 $\ddot{A}_l + 2(2l+1)(l-1)\,Oh\,\dot{A}_l + l(l+2)(l-1)A_l + lB_l = 0$

Variables with applied Characteristic Scales	$A_l' = R A_l$	$\dot{A}_l' = \frac{R}{t_\sigma} A_l$	$\ddot{A}_l' = \frac{R}{t_\sigma^2} A_l$	$B_l' = \frac{R^2 \rho}{t_\sigma^2} B_l$

Reid (1960) (Notation Modified) $q^2 = \frac{\bar{\sigma} R^2}{\nu}$, $\bar{\sigma} = \frac{q^2 \nu}{R^2}$

Al is non-dimensionalized with respect to R, the chosen scale for length. Likewise, Al first is non-dimensionalized with respect to R / t_sigma, to represent the scale of the rate of change of radius. Similar scaling is taken for the second derivative of Al.

Bl is the mode amplitude of pressure. The shown scale above describes pressure in terms of radius, inertio capillary time and density Rho.

The definition of Sigma_hat as shown in Aguero (2026) within the ODE is explained by Reid (1960). q within the equation represents the roots of the transcendental equation mentioned prior. The use of this equation helps reduce the non-dimensionalized equation specified by Aguero (2026) in terms of q and Oh.

Non-dimensionalization and Rescaling

$\ddot{A}'_\ell + (\bar{\sigma}_a + \bar{\sigma}_b)\dot{A}'_\ell + (\bar{\sigma}_a \bar{\sigma}_b) A'_\ell + \frac{\ell}{\rho R} B'_\ell = 0$	Aguero (2026)
$\frac{R}{t_\sigma^2}\ddot{A}_\ell + (\bar{\sigma}_a + \bar{\sigma}_b)\frac{R}{t_\sigma}\dot{A}_\ell + (\bar{\sigma}_a \bar{\sigma}_b) R A_\ell + \frac{\ell}{\rho R}\frac{R^2 \rho}{t_\sigma^2} B_\ell = 0$	Substitute Scales
$\ddot{A}_\ell + (\bar{\sigma}_a + \bar{\sigma}_b) t_\sigma \dot{A}_\ell + (\bar{\sigma}_a \bar{\sigma}_b) t_\sigma^2 A_\ell + \ell B_\ell = 0$	divide by $\frac{R}{t_\sigma^2}$
$\ddot{A}_\ell + \left(\frac{q_a^2 \nu}{R^2} + \frac{q_b^2 \nu}{R^2}\right) t_\sigma \dot{A}_\ell + \left(\frac{q_a^2 \nu}{R^2}\right)\left(\frac{q_b^2 \nu}{R^2}\right) t_\sigma^2 A_\ell + \ell B_\ell = 0$	$\bar{\sigma} = \frac{q^2 \nu}{R^2}$
$\ddot{A}_\ell + (q_a^2 + q_b^2)\left(\frac{\nu}{R^2}\right) t_\sigma \dot{A}_\ell + (q_a^2 q_b^2)\left(\frac{\nu^2}{R^4}\right) t_\sigma^2 A_\ell + \ell B_\ell = 0$	Factorize
$\ddot{A}_\ell + (q_a^2 + q_b^2)\left(\frac{\nu \rho^{\frac{1}{2}}}{R^{\frac{1}{2}} \sigma^{\frac{1}{2}}}\right)\dot{A}_\ell + (q_a^2 q_b^2)\left(\frac{\nu^2 \rho}{R \sigma}\right) A_\ell + \ell B_\ell = 0$	$t_\sigma = \frac{\rho^{\frac{1}{2}} R^{\frac{3}{2}}}{\sigma^{\frac{1}{2}}}$
$\ddot{A}_\ell + (q_a^2 + q_b^2)\left(\frac{\mu}{\rho^{\frac{1}{2}} \sigma^{\frac{1}{2}} R^{\frac{1}{2}}}\right)\dot{A}_\ell + (q_a^2 q_b^2)\left(\frac{\mu^2}{\rho \sigma R}\right) A_\ell + \ell B_\ell = 0$	$\nu = \frac{\mu}{\rho}$
$\ddot{A}_\ell + (q_a^2 + q_b^2)\, Oh\, \dot{A}_\ell + (q_a^2 q_b^2)\, Oh^2 A_\ell + \ell B_\ell = 0$	$Oh = \frac{\mu}{\sqrt{\rho \sigma R}}$

$$\ddot{A}_\ell + (q_a^2 + q_b^2)\, Oh\, \dot{A}_\ell + (q_a^2 q_b^2)\, Oh^2 A_\ell + \ell B_\ell = 0$$

$\downarrow$ is an approximation of

$$\ddot{A}_\ell + 2(2\ell+1)(\ell-1)\, Oh\, \dot{A}_\ell + \ell(\ell+2)(\ell-1) A_\ell + \ell B_\ell = 0$$

Galbard (2025)
Eq 3.16

Non-dimensionalization and Rescaling

In order to make the transcendental equation itself compatible with the problem formulation, it too must undergo rescaling.

$$\frac{\alpha^4}{q^4} + 1 = \frac{2(l-1)}{q^2}\left[l + (l+1)\,\frac{q - 2lQ_{l+1/2}(q)}{q - 2Q_{l+1/2}(q)}\right],$$ Reid (1960) Eq. 19

$$Q_{l+1/2}(q) = J_{l+3/2}(q)/J_{l+1/2}(q)$$ Reid (1960) Eq. 20

$$\omega_l^2 = l(l+2)(l-1)\frac{\sigma}{\rho R^3}, \quad \alpha^2 = \frac{\omega_l R^2}{\nu}$$ Reid (1960) Eq. 3, Eq. 17

$$\alpha^2 = \frac{\omega_l R^2}{\nu}$$ Reid (1960) Eq. 17

$$\alpha^4 = \frac{\omega_l^2 R^4}{\nu^2}$$ Squared for convenience

$$\alpha^4 = \left(l(l+2)(l-1)\right)\frac{\sigma}{\rho R^3}\left(\frac{R^4}{\nu^2}\right)$$ Definition for ω_l^2 Reid (1960) Eq. 3

$$\alpha^4 = l(l+2)(l-1)\frac{R\sigma}{\rho\nu^2}$$ Expand

$$\alpha^4 = l(l+2)(l-1)\frac{R\sigma\rho^2}{\rho\mu^2}$$ $\nu = \frac{\mu}{\rho}$

$$\alpha^4 = l(l+2)(l-1)\frac{R\sigma\rho}{\mu^2}$$ Simplify

$$\alpha^4 = l(l+2)(l-1)\frac{1}{Oh^2}$$ $Oh = \frac{\mu}{\sqrt{R\sigma\rho}}$

$$\frac{\alpha^4}{q^4} + 1 = \frac{2(l-1)}{q^2}\left[l + (l+1)\left(\frac{q - 2lQ_{l+\frac{1}{2}}(q)}{q - 2Q_{l+\frac{1}{2}}(q)}\right)\right]$$ Reid (1960) Eq. 19

$$\frac{\alpha^4}{q^4} + 1 = \frac{2(l-1)}{q^2}\left[l + (l+1)\left(\frac{q - 2lQ(q)}{q - 2Q(q)}\right)\right]$$ Argument of Q Removed due to Redundancy

$$\alpha^4 + q^4 = 2q^2(l-1)\left[l + (l+1)\left(\frac{q - 2lQ(q)}{q - 2Q(q)}\right)\right]$$ Multiply by q^4 to isolate α^4

$$\frac{l(l+2)(l-1)}{Oh^2} + q^4 = 2q^2(l-1)\left[l + (l+1)\left(\frac{q - 2lQ(q)}{q - 2Q(q)}\right)\right]$$ $\alpha^4 = \frac{l(l+2)(l-1)}{Oh^2}$

Non-dimensionalization and Rescaling

The equations, now unified in notation and scaling are as follows:
Coefficients of $\dot{A}_\ell$ and A_ℓ are defined by Oh and q

$$\ddot{A}_\ell + (q_a^2 + q_b^2)\,Oh\,\dot{A}_\ell + (q_a^2 q_b^2)\,Oh^2 A_\ell + \ell B_\ell = 0$$

q_a and q_b are found by solving for the transcendental equation:

$$\frac{\ell(\ell+2)(\ell-1)}{Oh^2} + q^4 = 2q^2(\ell-1)\left[\ell + (\ell+1)\left(\frac{q - 2\ell\,Q(q)}{q - 2Q(q)}\right)\right]$$

Where q_a and q_b are the two roots that minimizes positive Re(q), where both the transcendental equation's value and the its roots in q are both complex.

Unfortunately, this scaling results in problems of its own. Specifically, the division by the Ohnesorge number in the transcendental equation results makes the solving of the equation for Oh = 0 impossible. The case of no viscosity, where Oh = 0 is precisely where the approximations in Gabbard et al (2025) and the found equation are identical, necessitating us to rescale in terms of q, such that the Oh = 0 case can be found.

$$\text{For} \quad q^2 = \frac{\bar{q}^2}{Oh}$$

$$\ddot{A}_\ell + (q_a^2 + q_b^2)\,Oh\,\dot{A}_\ell + (q_a^2 q_b^2)\,Oh^2 A_\ell + \ell B_\ell = 0$$

$$\ddot{A}_\ell + \left(\frac{\bar{q}_a^2}{Oh} + \frac{\bar{q}_b^2}{Oh}\right) Oh\,\dot{A}_\ell + \left(\frac{\bar{q}_a^2}{Oh}\,\frac{\bar{q}_b^2}{Oh}\right) Oh^2 A_\ell + \ell B_\ell = 0$$

$$\ddot{A}_\ell + (\bar{q}_a^2 + \bar{q}_b^2)\,\dot{A}_\ell + (\bar{q}_a^2 \bar{q}_b^2)\,A_\ell + \ell B_\ell = 0$$

Non-dimensionalization and Rescaling

For $q^2 = \dfrac{\bar{q}^2}{Oh}$

$$\frac{l(l+2)(l-1)}{Oh^2} + q^4 = 2q^2(l-1)\left[l + (l+1)\left(\frac{q - 2l\,Q(q)}{q - 2Q(q)}\right)\right]$$

$$\frac{l(l+2)(l-1)}{Oh^2} + \frac{\bar{q}^4}{Oh^2} = 2\,\frac{\bar{q}^2}{Oh}(l-1)\left[l + (l+1)\left(\frac{\frac{\bar{q}}{\sqrt{Oh}} - 2l\,Q\left(\frac{\bar{q}}{\sqrt{Oh}}\right)}{\frac{\bar{q}}{\sqrt{Oh}} - 2Q\left(\frac{\bar{q}}{\sqrt{Oh}}\right)}\right)\right]$$

$$l(l+2)(l-1) + \bar{q}^4 = 2Oh\,\bar{q}^2(l-1)\left[l + (l+1)\left(\frac{\frac{\bar{q}}{\sqrt{Oh}} - 2l\,Q\left(\frac{\bar{q}}{\sqrt{Oh}}\right)}{\frac{\bar{q}}{\sqrt{Oh}} - 2Q\left(\frac{\bar{q}}{\sqrt{Oh}}\right)}\right)\right]$$

As Oh $\rightarrow$ 0, $\dfrac{\bar{q}}{\sqrt{Oh}} \rightarrow +\infty$, $2Oh\,\bar{q}^2(l-1)\left[l + (l+1)\left(\dfrac{\bar{q} - \sqrt{Oh}\,2l\,Q\left(\frac{\bar{q}}{\sqrt{Oh}}\right)}{\bar{q} - \sqrt{Oh}\,2Q\left(\frac{\bar{q}}{\sqrt{Oh}}\right)}\right)\right] \rightarrow 0$

Assuming the limit and the actual value of the formulation are equivalent, at Oh = 0, the transcendental equation collapses to $l(l+2)(l-1) + \bar{q}^4 = 0$

The equations are now as follows:

Coefficients of $\dot{A}_l$ and A_l are defined by Oh and $\bar{q}$

$$\ddot{A}_l + \left(\bar{q}_a^2 + \bar{q}_b^2\right)\dot{A}_l + \left(\bar{q}_a^2\,\bar{q}_b^2\right)A_l + l B_l = 0$$

q_a and q_b are found by solving for the transcendental equation:

$$l(l+2)(l-1) + \bar{q}^4 = 0 \qquad Oh = 0$$

$$l(l+2)(l-1) + \bar{q}^4 = 2Oh\,\bar{q}^2(l-1)\left[l + (l+1)\left(\frac{\frac{\bar{q}}{\sqrt{Oh}} - 2l\,Q\left(\frac{\bar{q}}{\sqrt{Oh}}\right)}{\frac{\bar{q}}{\sqrt{Oh}} - 2Q\left(\frac{\bar{q}}{\sqrt{Oh}}\right)}\right)\right] \qquad Oh \neq 0$$

Methods

For a continuation method, where each subsequent iteration uses the previous iteration's result as the input to solve an incrementally different problem, a method of solving the transcendental equation is necessary, for a given Oh, L and input initial guess.

For the equation is transcendental, there is no way to analytically solve for values of q. Therefore, iterative methods are used. The multivariable newton method is used, where the real and imaginary part of the transcendental equation are two different equations to be solved, for two variables, the real and imaginary part of q.

Therefore, in each increment of the continuation method, the previous solution is used as the starting point of the newton method, for a shift in either Oh or L. The solution resulting from the newton method is used for as the starting point for the next increment.

In order to implement a continuation method, a reliable "seed" value is required which is close enough to a basin of attraction, such that the next iteration is likely to latch onto the desired branch of solutions. As mentioned before, the approximation at Oh = 0 is identical to that of the true equations, causing the values of q for Oh = 0 to be a good "seed" value.

$2(2l+1)(l-1)Oh$ approximates $q_a^2+q_b^2$ & $l(l+2)(l-1)$ approximates $q_a^2 q_b^2$

let $K_1 = 2(2l+1)(l-1)Oh$ such that $q_a^2+q_b^2 = K_1$

$K_2 = l(l+2)(l-1)$ $\quad q_a^2 q_b^2 = K_2$

Based on $x = \dfrac{-b \pm \sqrt{b^2-4ac}}{2a}$ for $ax^2+bx+c=0$

$$q_a^2 = \frac{K_1 + \sqrt{K_1^2-4K_2}}{2} \qquad q_b^2 = \frac{K_1 - \sqrt{K_1^2-4K_2}}{2}$$

Methods

$$q_1 = \sqrt{\frac{K_1 + \sqrt{K_1^2 - 4K_2}}{2}} \qquad q_3 = \sqrt{\frac{K_1 - \sqrt{K_1^2 - 4K_2}}{2}}$$

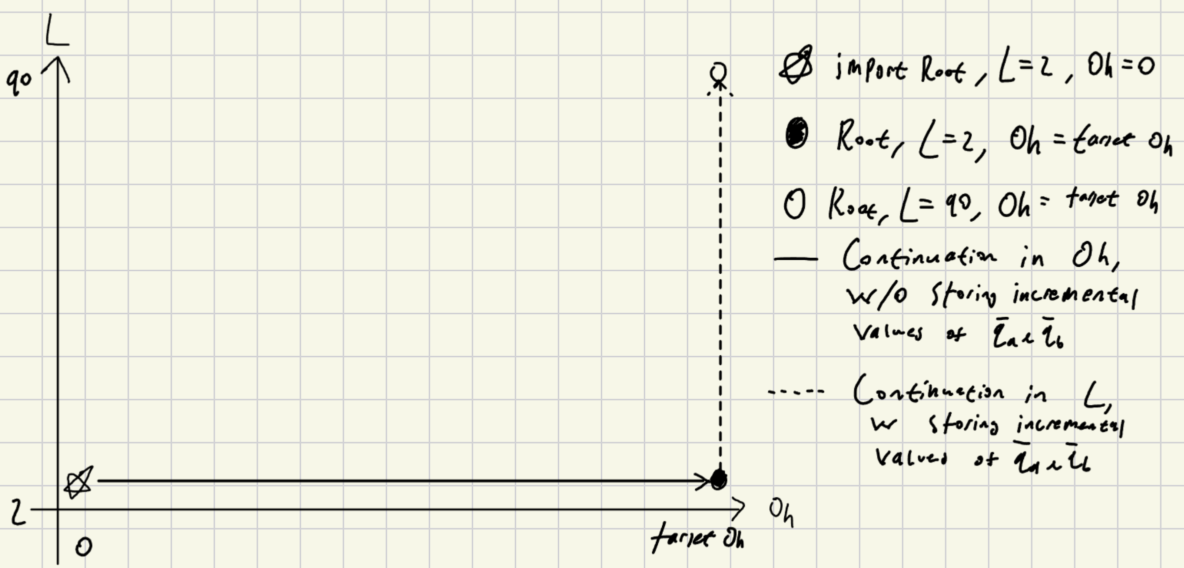
$$q_2 = -\sqrt{\frac{K_1 + \sqrt{K_1^2 - 4K_2}}{2}} \qquad q_4 = -\sqrt{\frac{K_1 - \sqrt{K_1^2 - 4K_2}}{2}}$$

This gives 4 candidate solutions, we name q1, q2, q3, q4

The 2 solutions out of these 4, for which Re(q) is smallest but positive, are the two values of q associated with the coefficients based on Gabbard et al (2025), for a given value of L and Oh

Each time the approximation from Gabbard et al (2025) is used as a starting point for the continuation method, we will refer to the process of calculating potential equivalent values of q and selecting the most appropriate roots as "importing" from Gabbard et al (2025)

Due to the nature of how the transcendental equation evolves as Oh and L are slowly increased, continuing along Oh first, then continuing along L has proven as the approach that most ensures that the roots remain along the branch of desired solutions.



Resources

Professor Rios' lecture on multivariable newton method:
https://drive.google.com/drive/folders/1Ab3lh--tZ3e7JE2tigNU_h__Lx6-FErs

Google Drive folder of all papers and resources cited, viscous drop notebooks and scripts:
https://drive.google.com/drive/folders/1L1cjmpMz3oNXWKHJusCcTDzezjjNyCQk

Repository of original Kinematic Match model script in Gabbard et al. (2025):
https://github.com/Katiekuehr/Drop_Simulations

Legend of Notational differences Across References

Quantity	Reid (1960)	Gabbard et al (2025)	Ragazzo (2019)	Aguero
Surface tension	T_1	σ	σ	σ
Principle mode oscillation frequency	$\sigma_{l:0}$	NA	ω_l	ω_l
Viscid mode oscillation frequency	$\sigma_{l:N}$	NA	$\tilde{\sigma}_{al}$ $\tilde{\sigma}_{bl}$	$\tilde{\sigma}_a$ $\tilde{\sigma}_b$
time-dependent amplitude	NA	A_l	A_l	A_l
1st derivative of time dependent amplitude	NA	U_l	$\dot{A}_l$	$\dot{A}_l$
2nd derivative of time dependent amplitude	NA	$\dot{U}_l$	$\ddot{A}_l$	$\ddot{A}_l$